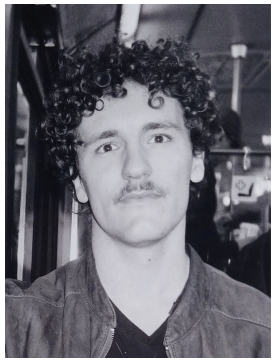
Protein Prediction II - CB + CS Exercise



Julien Schulz



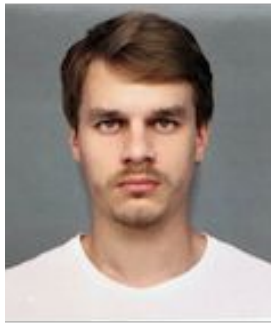
Tobias Senoner



Zeno Sewald



Julius Schlensok



Ivan Koludarov

https://www.cs.cit.tum.de/bio/home/

pp-org@rostlab.org

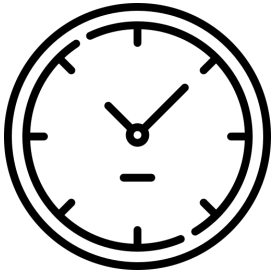
General Information

Organization - General

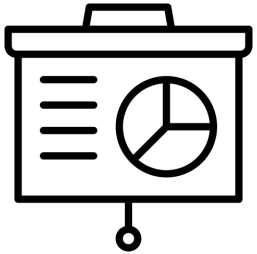


Monday

Room
5608.EG.038



10:15-11:45



Moodle Course



Project Work
(in groups)

Livestream / Recordings



Only for lectures
No recordings of project sessions

Project work

- ~4 students per Group
- Tri-weekly presentations starting from 02.11.26
- Feedback and questions as preparation for the exam



Exam Plan*



- Exam happens in project groups
- 15 min presentation about the project
- 10 min questions per student
- Grade bonus of 0.3

*This is an information and not the official announcement of the exam format.

Requirements

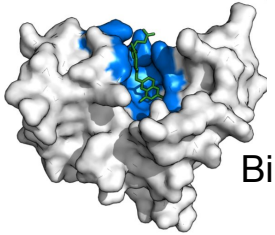
- Python Proficiency
- General Machine Learning Knowledge
- Self-Organization
- Motivation



Background

9

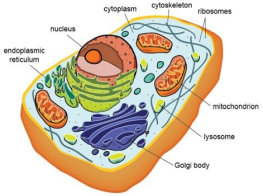
Who are we and what do we do?



Binding sites

Disorder/
Flexibility

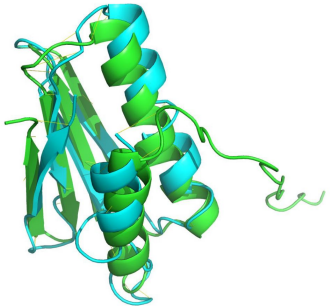


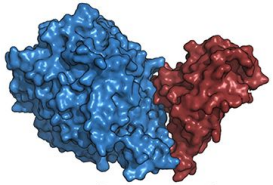


Function Prediction

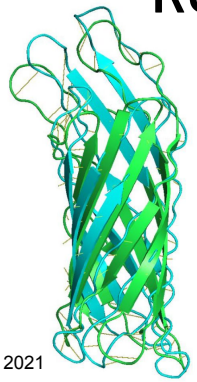
ROSTLAB

Structure Prediction

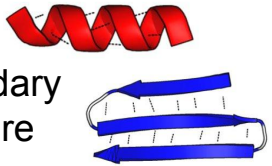




Protein-[Protein|DNA|RNA] interactions



Transmembrane regions



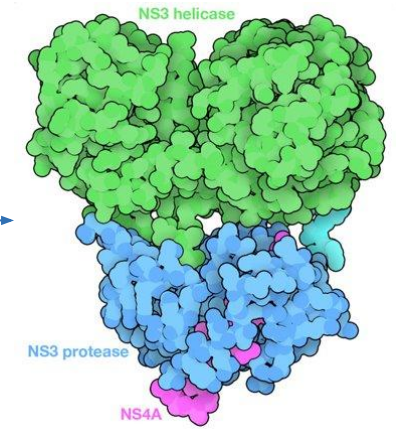
Secondary Structure

… and many more flavours

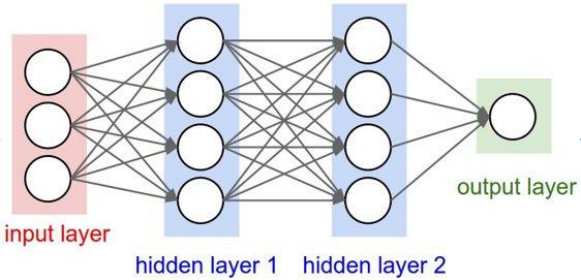
https://www.eurekalert.org/multimedia/pub/197685.php
https://upload.wikimedia.org/wikipedia/commons/e/ee/DHFR_methotrexate_inhibitor.png
https://www.frontiersin.org/files/Articles/261250/fmicb-08-01557-HTML/image_m/fmicb-08-01557-g001.jpg
https://www.creative-proteomics.com/services/subcellular-localization-service.htm
https://en.wikipedia.org/wiki/File:Alpha_beta_structure_(1).png
https://en.wikipedia.org/wiki/File:Alpha_beta_structure_(2).png

How do we predict?

PDB "Molecule of the month":
Hepatitis C Virus Protease/Helicase



SEQVENCE



Input Protein Sequence

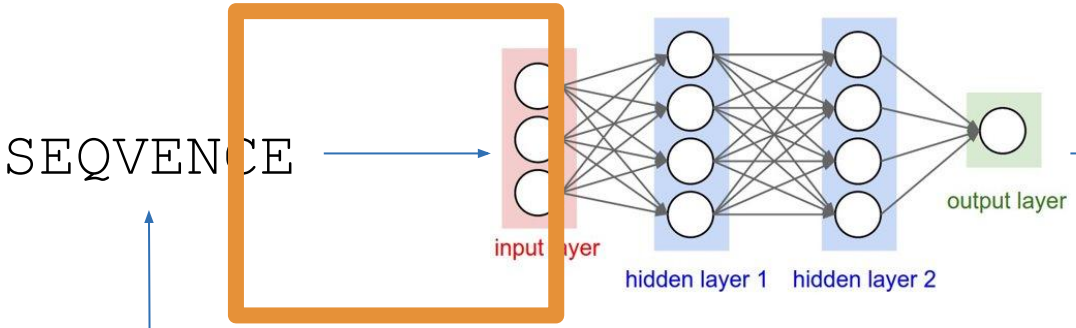
Overcome sequence-structure gap by predictions

Some output: e.g. structure

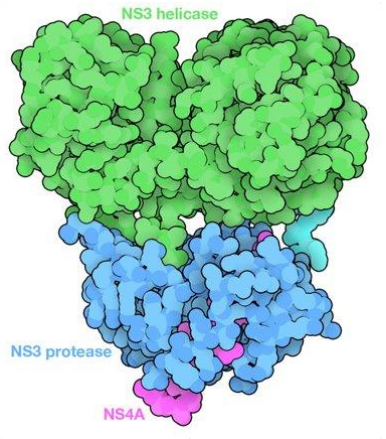
Burkhard Rost and Chris Sander. Bridging the protein sequence-structure gap by structure predictions. Annual Review of Biophysics and Biomolecular Structure, 25:113–136, 1996

~~How do we predict?~~

How do we represent proteins?



PDB "Molecule of the month":
Hepatitis C Virus Protease/Helicase



Input Protein Sequence

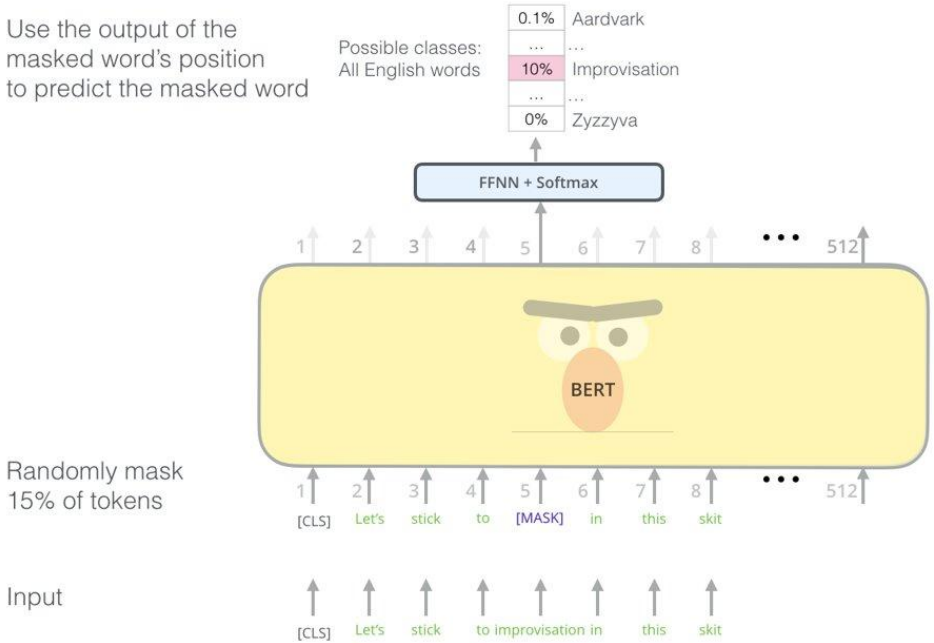
Overcome sequence-structure gap by predictions

Some output: e.g. structure

Burkhard Rost and Chris Sander. Bridging the protein sequence-structure gap by structure predictions. Annual Review of Biophysics and Biomolecular Structure, 25:113–136, 1996

MLM approaches: Transformer (BERT, T5, …)

Masked Language Modeling (MLM), i.e. reconstruct corrupted input tokens given all other tokens



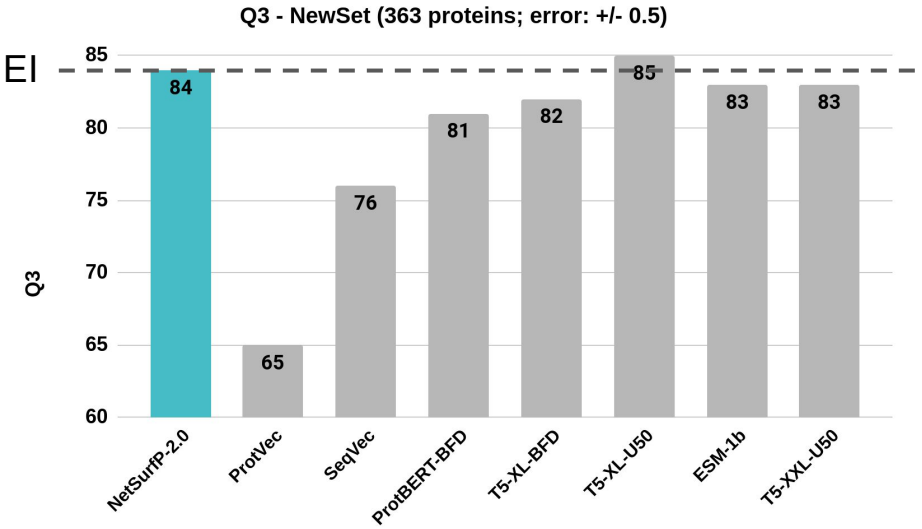
https://jalammar.github.io/illustrated-bert/

https://jalammar.github.io/illustrated-transformer/

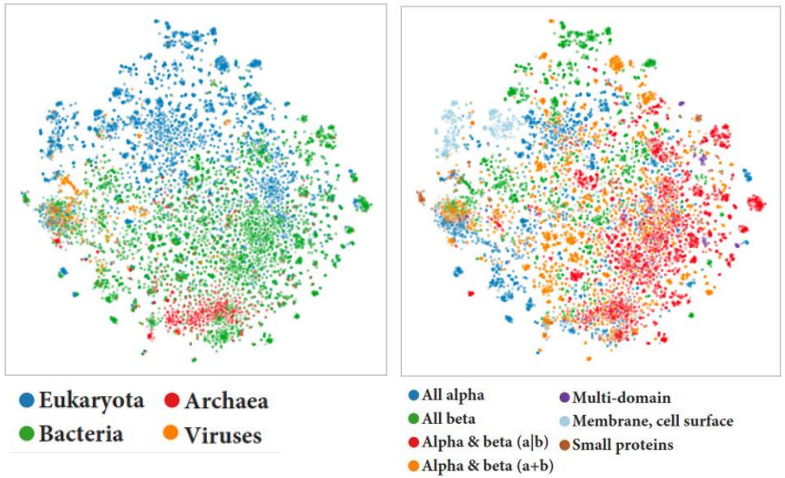
Vasawani A, et al. "Attention is all you need". (2017)

Devlin J., et al. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding". (2018)

Protein Language models (pLM) - ProtT5



Supervised training on labeled data



Unsupervised projection of labeled dataset

Elnaggar Ahmed, Heinzinger Michael, Dallago Christian, Rost Burkhard, et al. "ProtTrans: Towards cracking the language of life's code through self-supervised deep learning and high-performance computing" bioarxiv (2020)

Presentations

Minimum requirements

- Time limit: 10 min talk + 5 min questions/feedback
- Introduce yourself and your topic
- Answer questions in the presentation
  - What did you work on, and what element of the project does it tackle
  - What is the biggest success you have had since the last presentation
  - What is the current road-block you are facing
  - What are your next steps, why, and how does it help achieve the goal of the project?
  - How did you evaluate your current progress
- Any questions, discussion points, or findings you want to share

Key Principles

Clarity and Simplicity

- Use clear, concise language.
- Abstract away from details

Logical Structure

- Begin with a brief recap of objectives.
- Present results in a logical sequence.



Key Principles



Contextualization

- Explain how these results fit into the overall project.
- Highlight their significance.

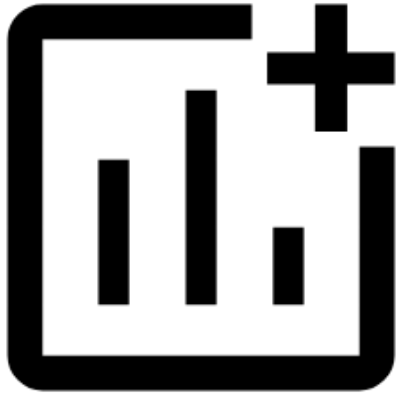
Interpretation

- Discuss what the results mean.
- Address any unexpected findings.

Key Principles

Effective Visualization

- Use graphs, charts, or tables where appropriate.
- Ensure visuals are easy to read and interpret.



Projects

20

Project 1: Pseudogene classification

What are pseudogenes?

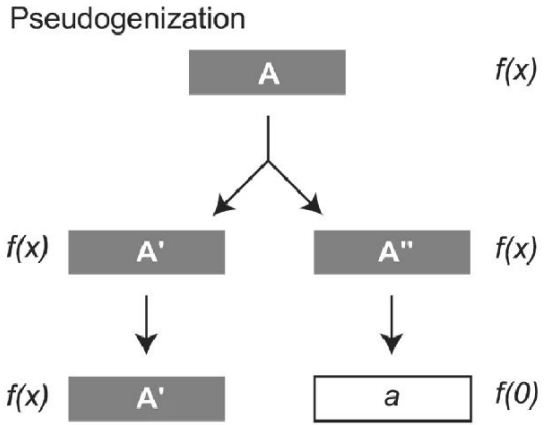
- ancient gene copies that no longer encode proteins

Why do we care?

- They resemble real proteins and sneak into databases by mistake

What do we want to learn?

1) Can pLMs [1] separate functional proteins from real-world pseudogene sequences?
(particularly on hard cases where the two are nearly identical at the sequence level)
2) How many mutations does it take until the model picks up on the degradation?



Visualization of the pseudogenization process (simplified). [2]

[1] Elnaggar, … & Rost, “ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning”, IEEE TPAMI (2022)
[2] Kramer, “Understanding the Genetic Basis of Floral Diversity”, BioScience (2007)

Project 2: Isoform plausibility oracle

What are isoforms?

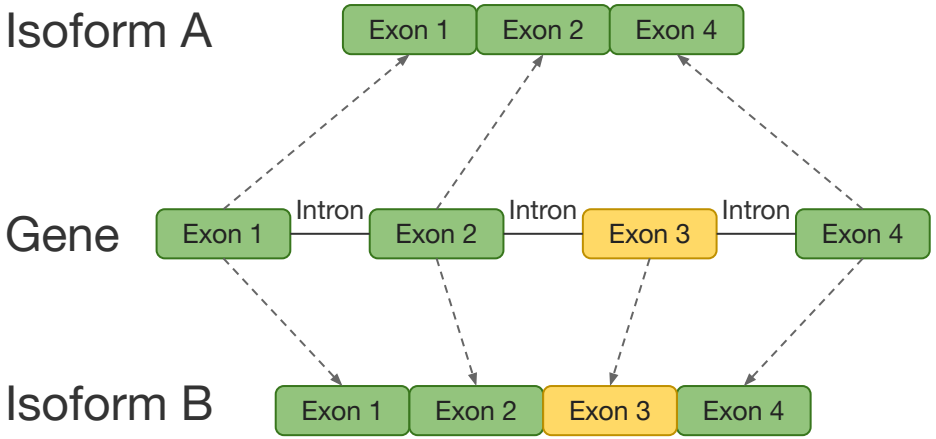
- different protein variants from the same gene (mainly via alternative splicing)

Why do we care?

- Many annotated isoforms are probably non-functional fragments
- Databases sometimes designate the wrong isoform as the main isoform

What do we want to learn?

1) Can pLMs [1] identify the main isoform of a gene?
2) Do pLM scores agree with the structure-based pLDDT scores [2] used by Sommer et al. [3]? Where they disagree with the MANE database [4] annotations: which one looks right?



Alternative splicing (here: exon skipping) leading to multiple isoforms

[1] Elnaggar, … & Rost, “ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning”, IEEE TPAMI (2022)
[2] Jumper, … & Hassabis, "Highly accurate protein structure prediction with AlphaFold", Nature (2021)
[3] Sommer, … & Salzberg, "Structure-guided isoform identification for the human transcriptome", eLife (2022)
[4] Morales, … & Murphy, “A joint NCBI and EMBL-EBI transcript set for clinical genomics and research”, Nature (2022)

Project 3: Predicting (un)biological database metadata

Why?

- Some database labels reflect database history more than biology
- Other transfer learning tasks might “cheat” using non-biological features

Example targets with decreasing biological meaning

- UniRef cluster size
- Has experimental structure?
- Is reviewed (Swiss-Prot vs. TrEMBL)?
- Date of entry
- …

Questions

Are certain database artifacts predictable using pLMs [1] embeddings & simple probes?
Is predictability explained by a simple baseline predictor using simple protein properties (length, AA composition, taxonomy, ...) instead of sequence/embeddings?

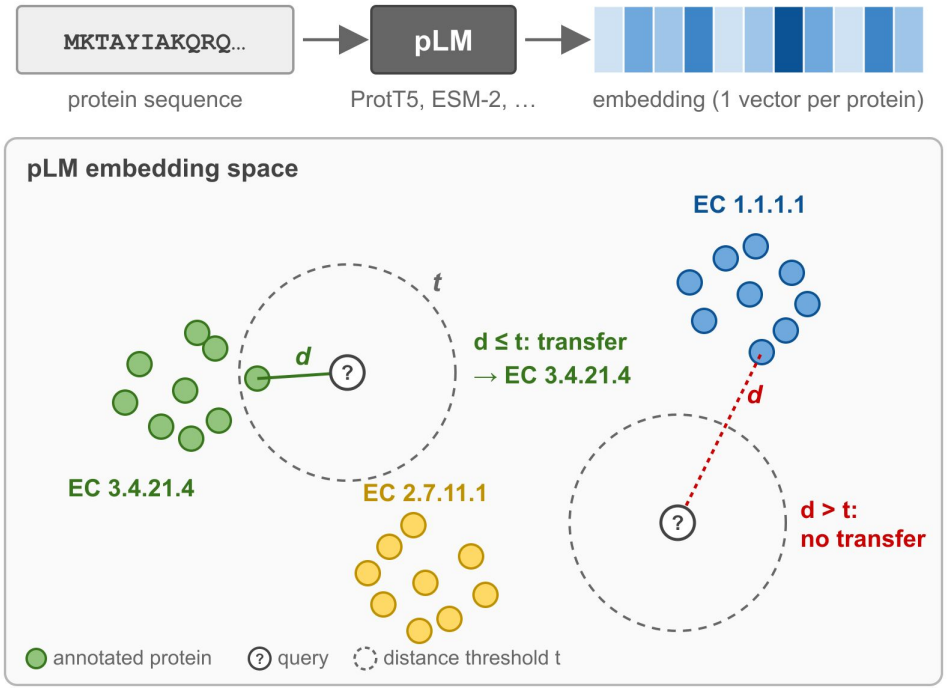
[1] Elnaggar, … & Rost, “ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning”, IEEE TPAMI (2022)

Project 4: Embedding-based annotation transfer (EAT)

What is EAT?
transfer annotations via nearest neighbours in pLM [1] embedding space [2]

Why do we care?
- training-free, yet beats sequence similarity for distant relatives [2, 3]
- but: thresholds and reliability scores stem from 2021-era pLMs

What do we want to learn?
1. Which pLM and distance (Euclidean vs. cosine) work best for structure (CATH) and function (EC)?
2. Up to which distance is transfer safe? → thresholds + updated reliability score

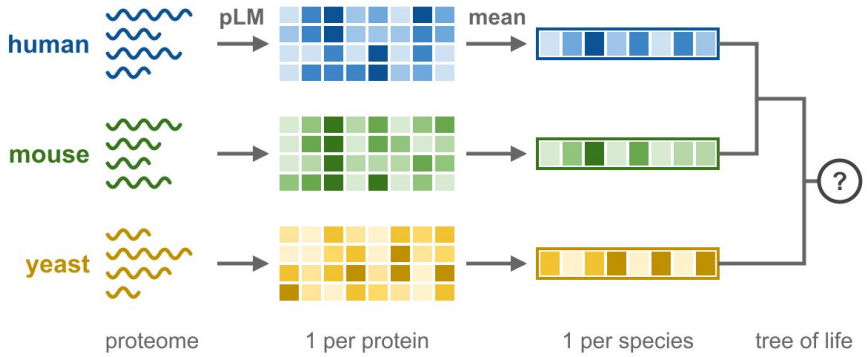


[1] Elnaggar, … & Rost, "ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning", IEEE TPAMI (2022)
[2] Littmann, … & Rost, "Embeddings from deep learning transfer GO annotations beyond homology", Sci Rep (2021)
[3] Heinzinger, … & Rost, "Contrastive learning on protein embeddings enlightens midnight zone", NAR Genom Bioinform (2022)

Project 5: One vector per species

What is it?
pool pLM [1] embeddings over a whole proteome into one vector per species
Why do we care?
– species “spectra” of predicted protein location recover deep clades, but not close species [2, 3]
– learned pooling of genome-wide pLM embeddings already predicts bacterial pathogenicity [5]; does a simple pooled vector carry the species itself, across all life?
What do we want to learn?

(1) Are proteins closer within a species than across species (matched orthologs)?
(2) Plain vs family-centred mean: which recovers relationships, and does it beat amino-acid composition?
(3) Can it place held-out clades in TaxEmbed [4]? (close species: a dated tree)



[1] Elnaggar, … & Rost, “ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning”, IEEE TPAMI (2022)
[2] Marot-Lassauzaie, … & Rost, “Spectrum of Protein Location in Proteomes Captures Evolutionary Relationship Between Species”, J Mol Evol (2021)
[3] Krautheimer, MSc thesis, TUM (2023)
[4] Koludarov & Rost, “TaxEmbed: one hyperbolic embedding for all named cellular life”, in preparation (model on Hugging Face)
[5] Ferrer Florensa, …, Rost & Aarestrup, “… bacterial pathogenic capacity … with PathogenFinder2”, bioRxiv (2025)

Presentation resources

Videos

- https://youtu.be/HnlWiVs9P5o?si=6NWPcEl5ompN7rbN
- https://youtu.be/4pymfPHQ6SA?si=VddVoJ8fwxIGwdo1
- https://youtu.be/Iwpi1Lm6dFo?si=-wrvsqleYEQwLBM9

Books

- “Presentation Zen” by Garr Reynolds
- “Designing Science Presentations” by Matt Carter
- “Better Data Visualizations” by Jonathan Schwabish

Other

- https://www.annualreviews.org/content/journals/10.1146/annurev-biodatasci-080917-013424

Background resources

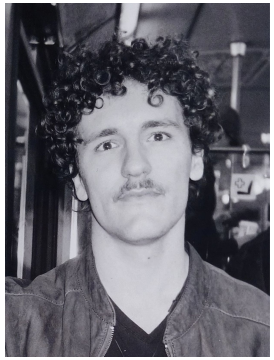
- **Introduction to Machine Learning:**
  - https://towardsdatascience.com/machine-learning-an-introduction-23b84d51e6d0
  - D Chicco. Ten quick tips for machine learning in computational biology. *BioData Mining*, 2017. https://doi.org/10.1186/s13040-017-0155-3
  - P Larranaga et al. Machine learning in bioinformatics. *Briefings in Bioinformatics*, 2006. https://doi.org/10.1093/bib/bbk007
- **Intro to transformers:** https://huggingface.co/course/chapter1/4?fw=pt
- **All about language models:** https://jalammar.github.io/
- **ProtTrans:** A Elnaggar, M Heinzinger et al. ProtTrans: Towards Cracking the Language of Life's Code Through Self-Supervised Deep Learning and High Performance Computing. *bioRxiv*, 2020. https://doi.org/10.1101/2020.07.12.199554
- **ESM-2**: Lin, Zeming, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Allan dos Santos Costa et al. "Language models of protein sequences at the scale of evolution enable accurate structure prediction." bioRxiv (2022). https://www.biorxiv.org/content/10.1101/2022.07.20.500902v2

Questions?



Thank you!

pp-org@rostlab.org








Group Assignment

- Please find a group of 4-5 people
- Find a Project you would like to work on together
- Discuss with the other people in the room that you aren't clashing -> each project can only be assigned once